

BLOCKCHAIN-ENABLED PAPER TRADING PLATFORM

for the Indian Stock Market

Real-Time Market Data	Virtual Trading	Blockchain Ledger	NSE / BSE	Smart Contracts
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Document Type	Project Overview & Abstract
Domain	FinTech · Blockchain · Stock Market Simulation
Market Scope	NSE & BSE — Indian Equity Markets
Core Technologies	Blockchain, WebSockets, React, Node.js, Solidity

ABSTRACT

The Indian stock market, comprising the National Stock Exchange (NSE) and the Bombay Stock Exchange (BSE), represents one of the fastest-growing financial ecosystems globally. However, a significant barrier for retail investors and students remains the steep learning curve associated with trading — particularly the financial risk of real-money experimentation. This project proposes the design and development of a **Blockchain-Enabled Paper Trading Platform** tailored for the Indian equity markets, where users can engage in simulated trading using virtual capital while experiencing real-time market data, live price feeds, order books, and actual market volatility. The platform functions as a full-featured simulated brokerage, providing users with a fixed virtual portfolio balance and a defined set of trades. Users can place market orders, limit orders, and stop-loss orders on NSE/BSE-listed securities through a responsive web interface powered by live WebSocket data feeds. The system replicates the actual market experience — including intraday fluctuations, circuit breakers, and after-market order queuing — without any real financial exposure. The defining innovation of this platform lies in the integration of **blockchain technology** as the underlying ledger for trade record-keeping. Every trade executed by a user is immutably recorded on a private or consortium blockchain (Hyperledger Fabric or Ethereum-compatible network), creating a tamper-proof, auditable, and transparent trade history. Smart contracts govern the virtual settlement logic — enforcing trade validation rules, computing profit and loss (P&L;), and updating virtual holdings — all without the possibility of manual tampering or data manipulation. Beyond individual trading simulation, the platform incorporates a **performance analytics engine** and a **leaderboard system**, enabling competitive learning environments for educational institutions, investor awareness programmes, and financial literacy initiatives aligned with SEBI's goals. The blockchain-anchored

records also serve as verifiable proof-of-skill credentials — a user's trading history, win rate, and return metrics can be cryptographically verified, offering a novel approach to financial talent discovery. This paper presents a comprehensive overview of the system architecture, the technology stack, the role of blockchain at each layer, and the design principles guiding the platform — contributing a scalable, trustworthy, and pedagogically valuable tool for the next generation of Indian retail investors.

Keywords: Paper Trading, Blockchain, Indian Stock Market, NSE, BSE, Smart Contracts, Hyperledger Fabric, Virtual Brokerage, Real-Time Data, SEBI, FinTech, Simulated Trading, WebSocket, DLT

1. INTRODUCTION & PROBLEM STATEMENT

Retail participation in Indian stock markets has surged post-COVID, with SEBI reporting over 100 million registered investors. Yet the dropout rate among new investors remains high due to inadequate preparation, impulsive trading decisions, and the psychological shock of real financial losses during the learning phase.

Traditional educational approaches — textbooks, theoretical courses — fail to replicate the emotional and analytical pressures of live trading. Existing paper trading platforms either lack real-time data fidelity, have no verifiable trade records, or are not localized for Indian market structures (NSE/BSE circuits, Indian trading hours, Rupee-denominated instruments).

This platform bridges that gap by delivering a **real-time, blockchain-backed, India-specific paper trading simulator** that is as close to the real experience as possible — without the financial risk.

2. PLATFORM OVERVIEW

2.1 User Experience Layer

Upon registration, each user receives a fixed virtual capital allocation (e.g., ₹5,00,000) and a defined number of trades (e.g., 50 trades per month). The dashboard mirrors a professional trading terminal:

- Live price ticker for NSE/BSE-listed equities, indices (NIFTY 50, SENSEX), and F&O instruments
- Real-time candlestick charts with technical indicators (RSI, MACD, Bollinger Bands)
- Order placement panel — Market Order, Limit Order, Stop-Loss, GTT (Good Till Triggered)
- Live order book depth (bid/ask spread simulation using real NSE data)
- Portfolio dashboard — Holdings, P&L, unrealized gains, day change
- Trade history log with timestamps, executed prices, and trade IDs

2.2 Simulated Brokerage Features

The backend replicates brokerage-level functionality:

- Brokerage fee simulation (e.g., 0.03% per trade — mirroring Zerodha/Upstox models)
- STT (Securities Transaction Tax), GST, and SEBI charges applied on virtual trades
- Margin trading simulation for F&O (Futures & Options) instruments
- Intraday vs. delivery position tracking and auto-square-off at 3:15 PM IST
- Circuit breaker enforcement — trades blocked when stocks hit upper/lower circuits
- Leaderboard with rank, total return %, Sharpe ratio, and trade accuracy

3. BLOCKCHAIN ARCHITECTURE & ROLE

3.1 Why Blockchain?

The core value proposition of blockchain in this platform is **immutability and verifiability**. In a conventional database, trade records can be altered, deleted, or manipulated. By anchoring every trade to a distributed ledger, the platform guarantees:

- Tamper-proof trade history — once written, a trade record cannot be modified

- Cryptographic proof of trade authenticity — each record is signed by the user's wallet key
- Transparent audit trail — institutions or recruiters can independently verify a user's trading history
- Decentralized P&L; settlement — smart contracts autonomously calculate and update balances

3.2 Blockchain Network Choice

Private/Consortium	Hyperledger Fabric	Enterprise-grade, permissioned, fast finality — Recommended
Public Testnet	Ethereum (Sepolia) + Solidity	Transparency, open verifiability, DeFi compatibility
Layer 2	Polygon (MATIC)	Low gas fees, Ethereum compatibility, scalability

3.3 Smart Contract Functions

Smart contracts (written in Solidity for Ethereum/Polygon or Go chaincode for Hyperledger) handle:

- **TradeExecute()** — Validates trade parameters, deducts virtual balance, mints a trade NFT/record
- **SettleTrade()** — On trade close, calculates net P&L;, updates holdings, emits event log
- **GetPortfolio()** — Read-only function returning current holdings and balance state
- **Leaderboard()** — Aggregates on-chain performance metrics for ranking computation
- **VerifyTrade(txHash)** — Public function allowing external verification of any trade record

3.4 On-Chain Trade Record Structure

Every executed trade is stored as a blockchain transaction containing:

Field	Type	Description
tradeId	bytes32	Unique cryptographic hash of the trade
userId	address	User's blockchain wallet address
symbol	string	NSE/BSE stock symbol (e.g., RELIANCE, INFY)
orderType	enum	BUY / SELL / SHORT / COVER
quantity	uint256	Number of shares
executedPrice	uint256	Price in paise (INR × 100) at execution
timestamp	uint256	Unix timestamp of execution
txHash	bytes32	Blockchain transaction hash for verification

4. COMPLETE TECHNOLOGY STACK

4.1 Frontend

- **React.js + TypeScript** — Component-based UI for trading terminal, charts, and dashboard
- **TradingView Lightweight Charts / Recharts** — Candlestick charts with real-time updates
- **Socket.io Client** — WebSocket connection for live price streaming
- **Redux Toolkit** — Global state management for portfolio, orders, and market data
- **Tailwind CSS** — Responsive, dark-mode trading UI
- **Web3.js / Ethers.js** — Wallet integration (MetaMask) for blockchain interaction

4.2 Backend

- **Node.js + Express.js** — RESTful API server for order management, user auth, and portfolio
- **Socket.io Server** — WebSocket server broadcasting real-time price updates
- **Python (FastAPI)** — Data processing service for technical indicator computation
- **Redis** — In-memory caching for live price data, reducing API latency
- **Bull Queue (Redis-based)** — Asynchronous job queue for trade processing and blockchain writes

4.3 Real-Time Market Data

- **NSE India WebSocket API / Upstox Market Data API** — Live tick data for NSE/BSE
- **Angel One SmartAPI / Fyers API** — Alternative real-time data providers (SEBI-regulated)
- **Yahoo Finance API (yfinance)** — Historical OHLCV data for backtesting and chart history
- **NSE Official Data** — EOD (End-of-Day) data, corporate actions, index composition

4.4 Blockchain Layer

- **Hyperledger Fabric** — Permissioned blockchain network for trade ledger (primary recommendation)
- **Solidity Smart Contracts** — Business logic for trade validation and P&L; settlement
- **IPFS (InterPlanetary File System)** — Decentralized storage for trade receipt documents
- **MetaMask / WalletConnect** — User wallet for signing trade transactions
- **Hardhat / Truffle** — Smart contract development, testing, and deployment framework
- **The Graph Protocol** — Indexing and querying blockchain trade history efficiently

4.5 Database

- **PostgreSQL** — Primary relational database for user accounts, order book, portfolio state
- **MongoDB** — Document store for trade history, market snapshots, and analytics data
- **Redis** — Session management, live price cache, leaderboard sorted sets
- **TimescaleDB** — Time-series extension for PostgreSQL to store tick-level price history

4.6 Infrastructure & DevOps

- **Docker + Docker Compose** — Containerized microservices architecture
- **AWS / GCP** — Cloud hosting (EC2/GKE for backend, RDS for database, S3 for static assets)
- **Nginx** — Reverse proxy, load balancing, WebSocket support
- **GitHub Actions / Jenkins** — CI/CD pipeline for automated testing and deployment

- **JWT + OAuth 2.0** — Secure authentication with Google/GitHub SSO support

5. SYSTEM ARCHITECTURE OVERVIEW

The platform follows a **microservices architecture** with distinct services communicating via REST APIs and message queues:

Layer	Service	Responsibility
Presentation	React Web App	Trading terminal UI, charts, portfolio dashboard
API Gateway	Node.js + Nginx	Route requests, auth middleware, rate limiting
Market Data	Python WebSocket Service	Consume NSE/BSE feeds, broadcast to clients
Order Engine	Node.js Order Service	Validate, match, and execute virtual orders
Blockchain Writer	Node.js + Web3	Write confirmed trades to blockchain ledger
Smart Contracts	Solidity / Go Chaincode	On-chain trade validation and P&L settlement
Analytics	Python FastAPI	Compute Sharpe ratio, win rate, technical indicators
Notification	Firebase / WebSocket	Real-time alerts for price targets, order fills

6. KEY FEATURES & DIFFERENTIATORS

Real-Time Market Fidelity	Tick-by-tick live data from NSE/BSE via SEBI-regulated APIs. Prices, volumes, and order book depth update in real-time — indistinguishable from live trading.
Blockchain Trade Ledger	Every trade is hash-anchored to a distributed ledger, creating an immutable, cryptographically signed record. Users can export a verifiable PDF certificate of their trading history.
Verifiable Skill Credentials	A user's return %, Sharpe ratio, and accuracy over defined periods are on-chain verifiable. Recruiters or institutions can validate trading performance without relying on self-reported data.
India-Specific Market Rules	Circuit breakers, T+1 settlement (new NSE rule), intraday square-off timings, STT/GST/SEBI charge calculation — all replicated accurately for the Indian market.
Competitive Leaderboards	Redis sorted sets + on-chain data power a real-time leaderboard. Institutions can create private leagues for students with customized virtual capital and trade limits.

Risk Analytics Dashboard

Integrated portfolio analytics: Max Drawdown, Beta, Volatility, CAGR, Sortino Ratio — helping users understand risk-adjusted performance beyond simple P&L;

7. REGULATORY & COMPLIANCE CONSIDERATIONS

Since the platform uses real market data and simulates brokerage operations, the following compliance aspects are addressed:

- **SEBI Compliance** — Platform clearly disclosed as a simulation; no real money involved; no investment advice provided
- **Data Licensing** — Real-time NSE/BSE data obtained through licensed API providers (Upstox, Angel One, Fyers)
- **KYC** — Basic identity verification for users; optional Aadhaar-based verification for institutional partnerships
- **Data Privacy** — User trading data stored securely; blockchain addresses pseudonymous; DPDP Act 2023 compliance
- **Blockchain Legality** — Using blockchain as a private ledger/database does not constitute cryptocurrency trading; fully legal in India

8. FUTURE ROADMAP

Phase 1	Core trading platform with NSE equities + blockchain ledger	Months 1–4
Phase 2	F&O (Futures & Options) simulation + margin trading	Months 5–7
Phase 3	Institution portal for colleges (SEBI Investor Awareness)	Months 8–10
Phase 4	Mobile app (React Native) + push notifications	Months 10–12
Phase 5	NFT trade certificates + DeFi integration experiments	Year 2
Phase 6	AI-powered trading assistant + strategy backtester	Year 2+

9. CONCLUSION

This Blockchain-Enabled Paper Trading Platform represents a convergence of three powerful trends: the democratization of financial education, the rise of retail participation in Indian equity markets, and the maturation of blockchain as a trustworthy record-keeping infrastructure.

By providing a real-time, high-fidelity trading simulation anchored by an immutable blockchain ledger, the platform does not merely teach trading — it **verifiably proves trading skill**. In an era where financial literacy is a national priority and SEBI actively promotes investor education, this platform offers a scalable, trustworthy, and technologically innovative solution.

The use of smart contracts for automated settlement, the pseudonymous yet verifiable nature of blockchain credentials, and the alignment with Indian market-specific regulations make this project both technically robust and practically impactful for millions of aspiring Indian investors.
